PV102 Day 5: Battery Storage Autonomy and Critical Loads

Students decide what the battery is for, then estimate storage need from critical loads, runtime, autonomy days, voltage, usable capacity, and safety boundaries.

PV102 | CTS Day 21 | Instructor Run Sheet
Storage has a defined job: carry named loads through named time. | Monday, August 10, 2026

Core rule

Storage has a defined job: carry named loads through named time.

Learning targets

- Explain the difference between battery capacity, usable capacity, and autonomy.
- Estimate storage need from essential daily Wh and autonomy goal.
- Identify why battery chemistry, BMS limits, temperature, ventilation, and overcurrent protection matter.
- Defend load shedding as a design tool, not a failure.

Readiness and materials

Before students arrive	Materials
Open slide deck and test display. Print student packets and one instructor key. Prepare three team stations. Review safety boundary before any equipment is handled.	Battery spec cards Load tables from Day 3 Calculators Scenario cards Storage worksheet Projector and slide deck

Five-hour run of show

Time	Block	Instructor move
10:00-10:15	Opening and continuity	Connect Day 4 to today's rule. Name what students already know and what new judgment is being added.
10:15-10:40	Morning quiz and review	Students work independently for 12-15 minutes. Review by asking for evidence, not just letters.
10:40-11:25	Concept lesson	Use slides 1-14. Pause after each technical rule and ask what decision it changes.
11:25-12:15	Guided practice	Model one example on the board. Students complete a quick pair-check before lunch.
12:15-12:45	Lunch/reset	Reset lab stations. Keep energized work out of scope unless explicitly authorized by instructor and equipment procedure.
12:45-1:00	Lab brief	Assign teams and roles. Review acceptance criteria and stop-work triggers.

1:00-2:15	Critical Load Storage Plan	Teams reduce an oversized wish list into a critical-load storage plan that a real system could support.
2:15-2:40	Handoffs and peer verification	Each team gives a concise technical handoff. Peers check evidence, boundary, and unknowns.
2:40-3:00	Exit ticket and reset	Collect packets. Preview next step: PV102 Day 6 connects storage and array decisions to inverter, charge controller, distribution, and protection boundaries.

Instructor talk track

Opening script: "PV102 is not about throwing parts at a problem until reality files a complaint. Today we use battery storage autonomy and critical loads to make the next design decision safer and more defensible. The rule is simple: Storage has a defined job: carry named loads through named time."

Teaching notes

Move	Look-for
Ask	What decision does this evidence change?
Listen for	Students separating facts, assumptions, and unknowns.
Correct gently	Component-first answers. Redirect to job, evidence, boundary, then component.
Require	A handoff that another person could act on without reading minds. Humanity keeps trying telepathy. It remains a poor project-management system.

Lab instructor procedure

- Start with the Day 3 load table or provided scenario.
- Mark each load as critical, useful, or optional.
- Calculate critical daily Wh.
- Choose an autonomy target and estimate required Wh before adjustment.
- List the battery information required before final sizing.
- Handoff the critical-load list, autonomy target, and rejected optional loads.

Answer snapshot and contingency plan

If equipment or printing is unavailable, run the lab with paper cards, chart paper, and oral handoffs. The learning target is decision quality, not hardware theater.

Safety and scope boundaries

No live electrical work is authorized by this lesson unless the instructor independently sets up a verified training system, follows manufacturer instructions, and establishes the required isolation, PPE, and supervision. Treat PV source circuits and batteries as energy sources, not props.

Day handoff

PV102 Day 6 connects storage and array decisions to inverter, charge controller, distribution, and protection boundaries.

Instructional sources

- U.S. Department of Energy, Solar Energy: https://www.energy.gov/topics/solar-energy
- U.S. Department of Energy, Solar Photovoltaic Technology Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-technology-basics
- U.S. Department of Energy, Solar Photovoltaic System Design Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-system-design-basics
- U.S. Department of Energy, Solar Integration: Solar Energy and Storage Basics: https://www.energy.gov/eere/solar/solar-integration-solar-energy-and-storage-basics
- U.S. Department of Energy, Solar Integration: Inverters and Grid Services Basics: https://www.energy.gov/eere/solar/solar-integration-inverters-and-grid-services-basics
- NREL PVWatts Calculator: https://pvwatts.nrel.gov/
- OSHA, Green Job Hazards - Solar Energy: Electrical: https://www.osha.gov/green-jobs/solar/electrical
- OSHA, Green Job Hazards - Solar Energy: Lockout/Tagout: https://www.osha.gov/green-jobs/solar/lockout-tagout
- OSHA, Green Job Hazards - Solar Energy: Personal Protective Equipment: https://www.osha.gov/green-jobs/solar/personal-protective-equipment
- Sean White, Solar Photovoltaic Basics, 2nd ed., Routledge, 2018, ISBN 9781138102866.

PV102 Day 5 Morning Quiz: Answer Key

10 questions | 15 minutes | Independent work

PV102 | CTS Day 21 | Morning Quiz - Answer Key
Storage has a defined job: carry named loads through named time. | Monday, August 10, 2026

Directions: Answer from your current knowledge. For written responses, use complete technical statements. If unsure, state the safe next action rather than guessing.

1. Battery capacity is commonly expressed as:

- A. Ah or Wh
- B. Degrees azimuth only
- C. Wire color
- D. Roof pitch only

Answer: A

Rationale/look-for: Battery capacity may be stated in amp-hours and/or watt-hours.

2. A battery bank should be sized from:

- A. Defined loads and runtime
- B. The prettiest battery case
- C. Random parts available
- D. Module color

Answer: A

Rationale/look-for: Storage follows the job it must perform.

3. Usable capacity is often less than nameplate capacity because:

- A. Depth-of-discharge and protection limits matter
- B. Labels are decoration
- C. All loads are AC
- D. Sunlight is heavy

Answer: A

Rationale/look-for: Chemistry and protection limits affect practical usable energy.

4. Autonomy means:

- A. Time the system must carry loads without enough solar input
- B. A compass direction
- C. A breaker type
- D. A module frame dimension

Answer: A

Rationale/look-for: Autonomy is stored-energy coverage during low or no production.

5. Load shedding means:

- A. Reducing or turning off lower-priority loads
- B. Cutting random wires
- C. Ignoring the client
- D. Making voltage disappear

Answer: A

Rationale/look-for: A prioritized load plan can protect critical function.

6. Estimate: 1,000 Wh/day for 2 days = ___ Wh before losses and usable-capacity adjustment.

_____

_____

Answer: 2,000 Wh

Rationale/look-for: Daily energy x autonomy days.

7. Why does battery voltage matter in system design?

_____

_____

Answer: Power/current/equipment match

Rationale/look-for: Higher system voltage reduces current for the same power and must match equipment.

8. Name one battery safety boundary.

_____

_____

Answer: Instructor judgment

Rationale/look-for: Accept OCPD, correct conductor size, terminals covered, BMS limits, ventilation as needed, temperature, no short circuits, manufacturer instructions.

9. Why is “more battery” not always the best answer?

_____

_____

Answer: Cost/risk/charge ability

Rationale/look-for: More storage costs more, occupies space, adds weight and protection needs, and must be rechargeable by the array.

10. Write the Day 5 rule in your own words.

_____

_____

Answer: Defined storage job

Rationale/look-for: Look for named loads and named runtime.

PV102 Day 5 Whiteboard Outline

PV102 | CTS Day 21 | Whiteboard Outline

Storage has a defined job: carry named loads through named time. | Monday, August 10, 2026

Board 1: Job	Board 2: Evidence	Board 3: Handoff
Topic: Battery Storage Autonomy and Critical Loads Rule: Storage has a defined job: carry named loads through named time. Question: What job are we solving?	Measured / Observed / Unknown Constraint that matters most Safety boundary before work	Decision Evidence Risk Unknowns Next action

Scenario anchors

- Bus conversion overnight loads: fridge, fan, lights, phone, laptop, water pump, optional induction cooktop.
- Remote sensor must survive two stormy days with no site visit.
- Small shop wants outage backup for lights, router, one outlet, and refrigerator only.

Final board sentence

Storage has a defined job: carry named loads through named time.

PV102 Day 5 Student Packet

Battery Storage Autonomy and Critical Loads

PV102 | CTS Day 21 | Student Print Packet
Storage has a defined job: carry named loads through named time. | Monday, August 10, 2026

Today's rule

Storage has a defined job: carry named loads through named time.

Learning targets

- Explain the difference between battery capacity, usable capacity, and autonomy.
- Estimate storage need from essential daily Wh and autonomy goal.
- Identify why battery chemistry, BMS limits, temperature, ventilation, and overcurrent protection matter.
- Defend load shedding as a design tool, not a failure.

Morning quiz

1. Battery capacity is commonly expressed as:

- A. Ah or Wh
- B. Degrees azimuth only
- C. Wire color
- D. Roof pitch only

2. A battery bank should be sized from:

- A. Defined loads and runtime
- B. The prettiest battery case
- C. Random parts available
- D. Module color

3. Usable capacity is often less than nameplate capacity because:

- A. Depth-of-discharge and protection limits matter
- B. Labels are decoration
- C. All loads are AC
- D. Sunlight is heavy

4. Autonomy means:

- A. Time the system must carry loads without enough solar input
- B. A compass direction
- C. A breaker type
- D. A module frame dimension

5. Load shedding means:

- A. Reducing or turning off lower-priority loads
- B. Cutting random wires
- C. Ignoring the client
- D. Making voltage disappear

6. Estimate: 1,000 Wh/day for 2 days = ___ Wh before losses and usable-capacity adjustment.

_____

_____

7. Why does battery voltage matter in system design?

_____

_____

8. Name one battery safety boundary.

_____

_____

9. Why is “more battery” not always the best answer?

_____

_____

10. Write the Day 5 rule in your own words.

_____

_____

Reference: decision order

Step	Action	Proof question
1	Define the job	What must the system accomplish?
2	Collect evidence	What is measured, observed, or still unknown?
3	Choose boundary	What must be protected, isolated, sized, or verified?
4	Defend the choice	What evidence supports the decision?
5	Handoff	What should the next person know?

Critical Load Storage Plan

Teams reduce an oversized wish list into a critical-load storage plan that a real system could support.

Scenario options

- Bus conversion overnight loads: fridge, fan, lights, phone, laptop, water pump, optional induction cooktop.
- Remote sensor must survive two stormy days with no site visit.
- Small shop wants outage backup for lights, router, one outlet, and refrigerator only.

Team procedure

- Start with the Day 3 load table or provided scenario.
- Mark each load as critical, useful, or optional.
- Calculate critical daily Wh.
- Choose an autonomy target and estimate required Wh before adjustment.
- List the battery information required before final sizing.
- Handoff the critical-load list, autonomy target, and rejected optional loads.

Team worksheet

Prompt	Team response
System/job statement	
Evidence used	
Decision made	
Safety boundary	
Top unknowns	
Next action	

Exit ticket

Which load did your team remove or limit, and why did that make the system stronger?

_____

_____

PV102 Day 5 Afternoon Lab

Critical Load Storage Plan

PV102 | CTS Day 21 | Afternoon Lab
Storage has a defined job: carry named loads through named time. | Monday, August 10, 2026

Mission

Teams reduce an oversized wish list into a critical-load storage plan that a real system could support.

Safety boundary

This lab is training work. Do not energize or connect equipment outside the instructor-approved procedure. When in doubt, stop, isolate, verify, and ask.

Scenario options

- Bus conversion overnight loads: fridge, fan, lights, phone, laptop, water pump, optional induction cooktop.
- Remote sensor must survive two stormy days with no site visit.
- Small shop wants outage backup for lights, router, one outlet, and refrigerator only.

Procedure

- Start with the Day 3 load table or provided scenario.
- Mark each load as critical, useful, or optional.
- Calculate critical daily Wh.
- Choose an autonomy target and estimate required Wh before adjustment.
- List the battery information required before final sizing.
- Handoff the critical-load list, autonomy target, and rejected optional loads.

Acceptance criteria

- Match the real job.
- Show the energy or work path.
- Name the dominant constraint.
- State the safety boundary.
- Record unknowns instead of guessing.
- Give a handoff another crew can use.

Team handoff form

Item	Response
Team members	
Scenario	
Primary decision	

Evidence	
Safety boundary	
Unknowns	
Next safest action	

PV102 Day 5 Classroom Post

Copy-ready announcement and assignment text

PV102 | CTS Day 21 | Google Classroom Post
Storage has a defined job: carry named loads through named time. | Monday, August 10, 2026

Announcement

Today in PV102 we are working on battery storage autonomy and critical loads. The core rule is: Storage has a defined job: carry named loads through named time.

Assignment title

PV102 Day 5 - Battery Storage Autonomy and Critical Loads

Student directions

- Complete the morning readiness quiz independently.
- Use the reference section during the afternoon lab.
- Work with your team to complete the scenario handoff.
- Complete the exit ticket before class ends.

Turn in

- Completed student packet
- Team lab handoff
- Exit ticket

Success criteria

Match the need. Show the path. Name the constraint. State the safety boundary. Record unknowns.